

Week 1 — The piles game (Bulgarian solitaire)

Facilitator guide for the three table adults

The session at a glance

The game. Counters sit in piles. A *move* takes one counter from every pile and puts those counters together as one new pile; piles that reach zero disappear. With 6 counters, $4, 1, 1 \rightarrow 3, 3 \rightarrow 2, 2, 2 \rightarrow 3, 1, 1, 1 \rightarrow 4, 2 \rightarrow 3, 2, 1$, and then $3, 2, 1 \rightarrow 3, 2, 1$ forever: the game *stops*. The theorem behind the session (Brandt, 1982): when the number of counters is a staircase number $1 + 2 + \cdots + k$ (3, 6, 10, 15, 21, ...), every start ends at the staircase $k, k - 1, \dots, 1$, which is the only arrangement that a move does not change. For any other number of counters the game never stops; it falls into a loop, and for some numbers (8 and 12 are the smallest) different starts can fall into different loops.

What the children are really investigating, in the order the pages ask it: what always happens (the same move, repeated, must stop or loop); which numbers of counters stop (existence and impossibility of a fixed point); whether every start ends in the same place (completeness: check every arrangement); how many moves it takes (optimality: the slowest start); which arrangements can never be reached (Garden of Eden states). The mathematics section at the end has the proofs; the answer sections have every run written out, so no adult needs to compute anything during the hour.

Shape of the hour: about 4 minutes of free handling of the counters, a 5-minute whole-group launch, about 40 minutes of table work, and 5 minutes of sharing at the end (54 minutes).

Notation in this guide: an arrangement is written biggest pile first with commas, 4, 3, 2, 1, and the children are asked to write their rows the same way. Each student packet has a core sequence that fills the 40 minutes, followed by extra problems on their own pages for a child or a table that is ahead; the table sections below say where the core ends and give the timings.

Materials and preparation

- Counters, one color is fine: 20 per child at the K–1 table (60), 15 per child at grades 2–3 (60), 21 per child at grades 4–5 (63), plus 6 large counters for the launch. About 190 in all; more is better, because children like to keep their finished staircases on the table. With 20 counters a K–1 child can keep the 3-, 6- and 10-stairs built at once (19 counters).
- The K–1 pictures are drawn for counters up to about 1.8 cm across. Larger counters still work: the child covers each circle with a counter and it does not matter if they overlap a little.
- Printed packets, one per child: K–1 (12 pages; the core is pages 1–6, Problems 1–7), grades 2–3 (9 pages; core pages 1–7, Problems 1–11), grades 4–5 (9 pages; core pages 1–6, Problems 1–12). Print one extra of each. Pencils, and a few blank sheets per table for runs that outgrow their tables.
- This guide for each adult. Read your table's section before the session; the runs are written out so you can check a child's record at a glance.
- Playing cards are not needed. If the 4–5 table wants to run a big game fast, a row of cards can stand for the piles (a 5 is a pile of 5, an ace is a pile of 1): a move replaces every card by the next lower card, removes the aces, and adds a card equal to the number of cards there were.
- Preparation: sort the counters into bags of 20, 15 or 21 per child, one bag per child, so that the launch needs no counting out. Put the K–1 children's chairs on the same side of the table as yours so you

can read the page and point.

Adults and tables

A suggestion, not a rule. The parent takes the K–1 table: the job there is reading aloud, pointing, keeping three sets of counters apart, and writing pile sizes when the game is too big to draw; every answer is in the K–1 section below, drawn as pictures, and the mathematics at that table is carried by the pictures. The second mathematician takes grades 2–3, where the map of all eleven arrangements (Problem 9) is the moment that needs someone who can ask “how do you know there is no other arrow?” The organizer takes grades 4–5, where the arguments (why the list of 11 arrangements is complete; why no arrangement of 8 counters can stand still; why some arrangements can never be reached) are the point. Around the 30-minute mark, the two mathematicians should each spend a minute at the other tables.

Free handling (about 4 minutes)

Give every child their bag of counters and nothing else. Let them make piles, towers, rows, and count. Two useful things to say, once, to anyone: “Can you make some piles so that no two piles are the same size?” and “How many counters are in your tallest tower?” Do not explain the move yet.

Whole-group launch (5 minutes)

The organizer stands where everyone can see, with 6 large counters.

1. “Here are 6 counters in three piles: 4, 1 and 1.” Make the piles on the table, visibly.
2. “Here is the only rule of today’s game. A move: I take one counter from every pile,” (do it: three counters in your hand, the two single piles have vanished, the pile of 4 is now 3) “and I put the ones I took together as a new pile.” Put them down: piles of 3 and 3.
3. “Everybody: make piles of 4, 1 and 1 with your own counters. Make one move.” Walk round; every table should have 3, 3. Fix any hand that took from only one pile. “Make another move.” Everyone should have 2, 2, 2. Check the tables again; this is the minute that saves ten wrong rows later.
4. “What happens next, you will find out at your tables. Two questions for the whole hour: does the game ever stop, and does it matter how you start? Your pages tell you what to try.”

Stop after the two moves; the grades 2–3 pages begin by recording this very game to its end. Say nothing about staircases, loops or the theorem. The launch is only the move.

K–1 table (three children, one adult)

Read each problem aloud, then read it again in pieces while the child acts. The child places counters on the picture (one counter per circle), makes the move on the table, and records the result in the next box. For the games with 3 to 6 counters the child draws: one dot per counter and a ring around each pile; the boxes are large enough for a ring of six dots. For the games with more than 6 counters (Problems 6, 12 and 13) the page says so: the child says the pile sizes aloud, biggest first, and you write them in the box (“9, 1”, “8, 2”); two boxes are the same when the numbers are the same. If a child does not want to draw at all, write the sizes for them in every problem and let the child draw the rings around your numbers. Keep each child’s counters on their own page or on a sheet of paper so the sets do not mix.

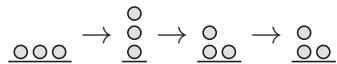
Order and timing (40 minutes): Problem 1 (5 min), 2 (6), 3 (5), 4 (5), 5 (6), 6 (8, you write), 7 (3). Problems 8–13, on the following pages, are for a child who is ahead; 12 and 13 need you to write.

Two rules for reading the boxes: a game has *stopped* when a box looks the same as the box just before it; a game goes *round and round* when a box looks the same as an earlier box that is not the one just before it. The stairs test (Problems 1, 3, 6, 8): put the piles down as towers, biggest on the left, next to the picture, then place the counters on the circles.

Answers, K–1

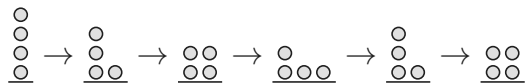
Each run is drawn below as the child should see it, box by box.

- **Problem 1** (3 counters, three piles of 1):



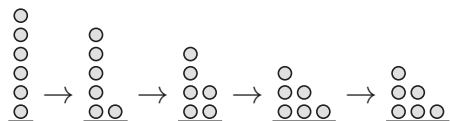
Boxes 2 and 3 are the same. Say: “The move did not change anything. The game stopped.” The three counters fit the stairs picture exactly: the game stopped at the stairs, 2 next to 1.

- **Problem 2** (4 counters, one pile):



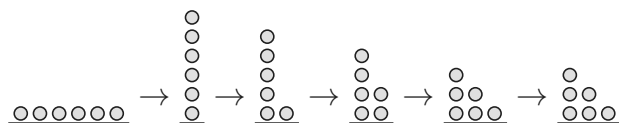
Box 4 is the same as box 1, box 5 the same as box 2. It never stops: the pictures 3, 1 then 2, 2 then 2, 1, 1 keep coming back in that order. Let the child make a few more moves and say “it goes round and round.”

- **Problem 3** (6 counters, one pile):



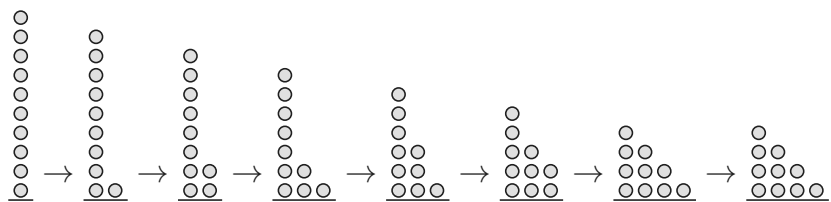
Box 4 is the same as box 3: it stops at 3, 2, 1 after 3 moves, and boxes 5 and 6 stay empty. The counters fit the 3, 2, 1 stairs exactly.

- **Problem 4** (6 counters, six piles of 1):



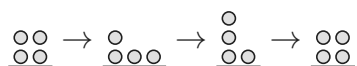
Box 5 is the same as box 4. The first box that shows the stairs is box 4, so 4 moves: count “one, two, three, four” along the number line while pointing at boxes 1 to 4. Notice with the child that after one move this is Problem 3 again.

- **Problem 5** (build the stairs): the stairs with 4 steps use 10 counters; with 5 steps, 15. A child who counts 3, 6, 10, 15 aloud has found the staircase numbers. Ask: “How many more counters did the bigger stairs need than the smaller one?” (3, then 4, then 5.)
- **Problem 6** (10 counters, one pile; you write the sizes):



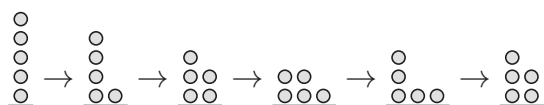
Box 7 is the same as box 6: it stops after 6 moves at 4, 3, 2, 1, and the counters fit the big stairs picture of Problem 5 exactly. Say each row with the child before writing it (“nine and one”).

- **Problem 7** (table): 3 stopped; 4 round and round; 6 stopped; 10 stopped. The two blank rows are for 5, 7 or 8 from Problems 9, 12 and 13 (all three go round and round). Ask “which numbers stopped?” and let the child say 3, 6, 10.
- **Problem 8** (4 counters, piles 2, 2):



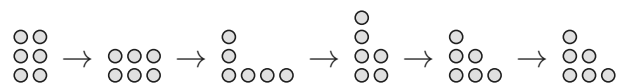
Box 3 is the same as the start. Four counters do not fit either stairs picture exactly: one counter is left over on the small stairs, and two circles stay empty on the big stairs. This is the first impossibility of the day: there are no stairs with 4 counters.

- **Problem 9** (5 counters, one pile):



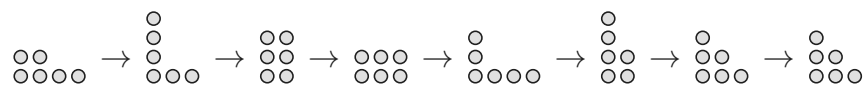
Boxes 2 and 5 are the same; the game goes round on the three pictures 3, 2 then 2, 2, 1 then 3, 1, 1 and never stops. In the table: 5, round and round.

- **Problem 10** (6 counters, piles 3, 3):



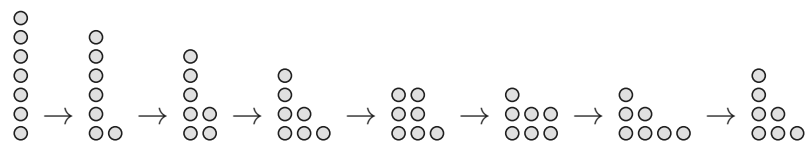
Box 5 is the same as box 4; the first stairs box is box 4, so 4 moves.

- **Problem 11** (6 counters, piles 2, 2, 1, 1):



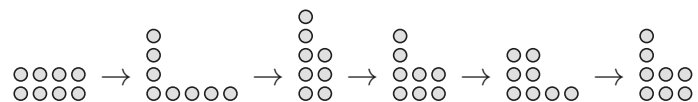
Box 7 is the same as box 6; 6 moves, the most of the four games with 6 counters (3, 4, 4, 6). All four stopped at the same stairs. Ask: “Is there any way to start 6 counters so that it does not end at the stairs?” Let them try one or two of their own; every start with 6 counters ends at 3, 2, 1 (the full list is in the tables at the end of this guide).

- **Problem 12** (7 counters, one pile; you write the sizes):



Box 7 is the same as box 3; the game goes round on four boxes and never stops. In the table: 7, round and round.

- **Problem 13** (8 counters, piles 2, 2, 2, 2; you write the sizes):



Boxes 3 and 5 are the same; this game goes round on only two boxes, 4, 2, 2 and 3, 3, 1, 1. In the table: 8, round and round.

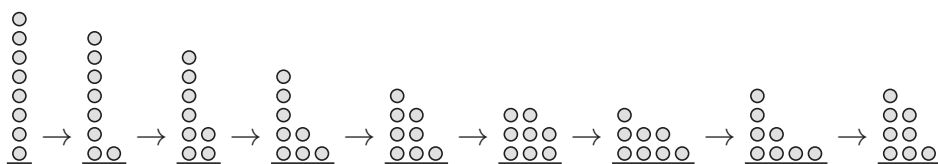
Hints, K–1, in the order to give them

1. A child takes counters from only some piles: “One from every pile. Point to each pile and take one.” Have the child put the taken counters in their other hand before making the new pile.
2. A child forgets that the counters taken form one pile: “All the ones in your hand go down together.”
3. A child cannot see whether two boxes are the same: count the piles in each box, then the counters in the biggest pile. Say the sizes aloud, “three and one”, for both boxes. If the drawings are too rough to compare, write the sizes under the dots yourself from now on.
4. Round and round is hard to see: put a finger on box 1 and ask “where is this picture again?”
5. Counting the moves to the stairs (Problems 4, 10, 11): point at box 1 and count along the number line until the first stairs box.
6. For the stairs test (Problems 1, 3, 6, 8): put the piles down as towers, biggest on the left, next to the picture, before placing counters on the circles.

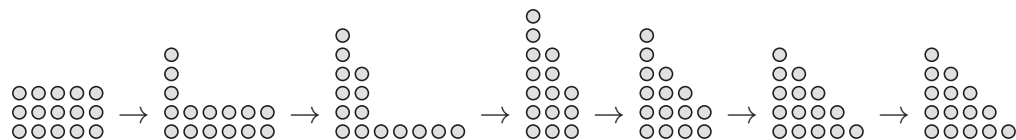
If the K–1 table stalls or finishes early

Stalls: skip to Problem 3 (the 6-counter game from one pile is short and satisfying), then come back. If drawing is slowing everything, you write the pile sizes and the child moves. If a child only wants to build, let them build the stairs of Problem 5 and count; the table of Problem 7 can be filled from Problems 1, 2, 3 and 6 alone.

Finishes early: Problems 8–13 in order (8 and 9 are quick; 10 and 11 answer “do all starts with 6 counters end at the stairs?”; 12 and 13 are the first games that never reach stairs and go round on four or two boxes). After that, on a blank sheet with boxes you draw: (1) 8 counters in one pile,



which goes round on four boxes, so with 8 counters the start of Problem 13 and this one go round on different pictures. (2) “Which numbers stopped?” 3, 6, 10: build all three stairs in a row and ask what the next one needs (15). (3) 15 counters as five piles of 3,



which stops after 5 moves at the 5-step stairs (you write the sizes; a child with 20 counters has enough).

Grades 2–3 table (four children, one adult)

Each child keeps their own counters and their own packet, but let pairs check each other's rows: a wrong row is usually a pile that was not reduced. In the tables, the children write pile sizes biggest first with commas, as the first problem shows ("3, 3"), so that equal arrangements look equal. The run tables have two or three more rows than the longest run needs, so the number of rows says nothing about the answer; a child who writes one wrong row still has room.

Order and timing (40 minutes): Problems 1–2 (7 min), 3–4 (6), 5–6 (6), 7–8 (6), 9, the map (7), 10 (4), 11 (4). Problem 3 is split: two children run 3, 3 and two run 2, 2, 1, 1, and they pool the results in Problem 4. Problems 12–15, on the last two pages, are for whoever is ahead; 14 can be shared out by the whole table.

Answers, grades 2–3

- **Problem 1** (4, 1, 1): 4, 1, 1 → 3, 3 → 2, 2, 2 → 3, 1, 1, 1 → 4, 2 → 3, 2, 1 → 3, 2, 1. The row for move 6 repeats move 5, so the game stopped after 5 moves at 3, 2, 1.
- **Problem 2:** 6 → 5, 1 → 4, 2 → 3, 2, 1 → 3, 2, 1 (stopped after 3 moves). 1, 1, 1, 1, 1, 1 → 6 → 5, 1 → 4, 2 → 3, 2, 1 → 3, 2, 1 (4 moves).
- **Problem 3:** 3, 3 → 2, 2, 2 → 3, 1, 1, 1 → 4, 2 → 3, 2, 1 → 3, 2, 1 (4 moves). 2, 2, 1, 1 → 4, 1, 1 → 3, 3 → 2, 2, 2 → 3, 1, 1, 1 → 4, 2 → 3, 2, 1 → 3, 2, 1 (6 moves).
- **Problem 4:** all five stopped. Moves: 4, 1, 1 takes 5; 6 takes 3; 1, 1, 1, 1, 1, 1 takes 4; 3, 3 takes 4; 2, 2, 1, 1 takes 6, the longest. All five end at 3, 2, 1, which drawn as towers is a staircase. Ask "did any start end somewhere else?" and then "could one?"; hold the second question until Problem 9.
- **Problem 5** (4): 4 → 3, 1 → 2, 2 → 2, 1, 1 → 3, 1 → 2, 2 → 2, 1, 1. Rows 1 and 4 are 3, 1; the game never stops. It cycles through 3, 1, 2, 2, 2, 1, 1.
- **Problem 6** (all five pictures of 4): A (4) → D; B (2, 2) → E; C (1, 1, 1, 1) → A; D (3, 1) → B; E (2, 1, 1) → D. The game goes round on D, B, E. The picture never reached by a move is C, four piles of 1: after any move there is a pile of size equal to the number of piles before the move, which is never 1 unless there was only one pile, and one pile of 4 becomes 3, 1, not 1, 1, 1, 1.
- **Problem 7** (7): 7 → 6, 1 → 5, 2 → 4, 2, 1 → 3, 3, 1 → 3, 2, 2 → 3, 2, 1, 1 → 4, 2, 1. Row 7 repeats row 3: it goes round, and the loop has four arrangements: 4, 2, 1, 3, 3, 1, 3, 2, 2, 3, 2, 1, 1.
- **Problem 8:** 1, 1, 1 → 3 → 2, 1 → 2, 1, stopped after 2 moves. Table: 3 stops; 4 goes round, 3 arrangements; 6 stops; 7 goes round, 4 arrangements. The blank row is for 5 (Problem 12: goes round, 3 arrangements), 10 (Problem 10: stops) or 8 (Problem 11: goes round, 4 or 2).
- **Problem 9** (the map of 6 counters). Arrows: A (2, 2, 1, 1) → B (4, 1, 1) → C (3, 3) → D (2, 2, 2) → I (3, 1, 1, 1) → H (4, 2) → K (3, 2, 1); E (1, 1, 1, 1, 1, 1) → F (6) → G (5, 1) → H; J (2, 1, 1, 1, 1) → G; K → K (the small loop arrow from K to itself). No arrow points at A, E or J. Every path ends at K. From A it is 6 arrows to K. Two arrows arrive at G (from F and J) and two at H (from G and I). Each picture has exactly one arrow leaving it, because the move leaves no choice; that is why the drawing is a tree of paths flowing into K, not a tangle. Since these are all the arrangements of 6 counters, the map is a complete proof that every start with 6 counters ends at the stairs.
- **Problem 10:** the next stairs, 4, 3, 2, 1, has 10 counters. Prediction: most children say "stops" because 3 and 6 stopped; some say 9 should be next ("every third number"); accept any prediction and test. 10 → 9, 1 → 8, 2 → 7, 2, 1 → 6, 3, 1 → 5, 3, 2 → 4, 3, 2, 1 → 4, 3, 2, 1: stops after 6 moves at the 4, 3, 2, 1 stairs.

- **Problem 11** (8 counters): $8 \rightarrow 7, 1 \rightarrow 6, 2 \rightarrow 5, 2, 1 \rightarrow 4, 3, 1 \rightarrow 3, 3, 2 \rightarrow 3, 2, 2, 1 \rightarrow 4, 2, 1, 1 \rightarrow 4, 3, 1$, row 8 repeats row 4: a loop of four (4, 3, 1, 3, 3, 2, 3, 2, 2, 1, 4, 2, 1, 1). $2, 2, 2, 2 \rightarrow 4, 1, 1, 1, 1 \rightarrow 5, 3 \rightarrow 4, 2, 2 \rightarrow 3, 3, 1, 1 \rightarrow 4, 2, 2$, row 5 repeats row 3: a loop of two (4, 2, 2, 3, 3, 1, 1). Both go round, and no: they go round on different arrangements. This is the first time in the session that the start matters for where you end up.
- **Problem 12** (5 and 1, 1, 1, 1, 1): $5 \rightarrow 4, 1 \rightarrow 3, 2 \rightarrow 2, 2, 1 \rightarrow 3, 1, 1 \rightarrow 3, 2$ (row 5 repeats row 2); $1, 1, 1, 1, 1 \rightarrow 5 \rightarrow 4, 1 \rightarrow 3, 2 \rightarrow 2, 2, 1 \rightarrow 3, 1, 1 \rightarrow 3, 2$ (row 6 repeats row 3). Yes, both go round on the same three arrangements 3, 2, 2, 2, 1, 3, 1, 1; the second game is the first one delayed by one move.
- **Problem 13**: $2, 2, 2, 2, 2 \rightarrow 5, 1, 1, 1, 1, 1 \rightarrow 6, 4 \rightarrow 5, 3, 2 \rightarrow 4, 3, 2, 1 \rightarrow 4, 3, 2, 1$ (stopped after 4 moves). $5, 5 \rightarrow 4, 4, 2 \rightarrow 3, 3, 3, 1 \rightarrow 4, 2, 2, 2 \rightarrow 4, 3, 1, 1, 1 \rightarrow 5, 3, 2 \rightarrow 4, 3, 2, 1 \rightarrow 4, 3, 2, 1$ (stopped after 6 moves). Both end at the stairs.
- **Problem 14** (all 22 arrangements of 8): the seven that go round on two are 6, 1, 1, 5, 3, 4, 2, 2, 4, 1, 1, 1, 1, 3, 3, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, 1. The other fifteen go round on four. Every run is written out in the table for 8 counters at the end of this guide.
- **Problem 15**: the 5-step stairs has 15 counters. $3, 3, 3, 3, 3 \rightarrow 5, 2, 2, 2, 2, 2 \rightarrow 6, 4, 1, 1, 1, 1, 1 \rightarrow 7, 5, 3 \rightarrow 6, 4, 3, 2 \rightarrow 5, 4, 3, 2, 1 \rightarrow 5, 4, 3, 2, 1$: stopped after 5 moves at the 5-step stairs.

Hints, grades 2–3, in the order to give them

1. Wrong row: “Count your piles before the move. That number is the size of the new pile.” Then “every old pile lost one.”
2. A child does not notice a repeat: “Read row 1 aloud. Now find it lower down.” If the match is the row just above, the game stopped; if it is higher up, it goes round.
3. Problem 6: “Make picture A with counters, make the move, then look for the picture you have.”
4. Problem 9: “Do the picture with the most piles first (E). Where does its arrow go?” Then “which pictures have you not drawn an arrow from yet?” Finally “are there pictures with no arrow coming in?”
5. After Problem 9, for the question “could a start with 6 counters end somewhere else?”: “Every arrangement of 6 counters is on this map. Where does each arrow chain end?”
6. Problem 10: “The stairs for 3 and 6 look like steps. Add one more step to the 3, 2, 1 stairs.”

If the grades 2–3 table stalls or finishes early

Stalls: go back to counters and one move at a time, one child moving while the others predict the next row aloud; then continue the table. If Problem 9 is too much, do it as a group with one map: each child takes three pictures. If the table is far behind at 25 minutes, go straight from Problem 6 to the map (Problem 9) and then Problem 10; Problems 7, 8 and 11 can be dropped.

Finishes early: Problems 12–15 in order (14 can be shared out across the table). After that: (1) “Find a start with 6 counters that takes more than 6 moves” (there is none; the map shows 6 is the most, from A). (2) “Predict the loop for 11 counters”: the loop for 7 counters is the 3, 2, 1 stairs with one extra counter placed on a step or as a new pile of 1 (4, 2, 1, 3, 3, 1, 3, 2, 2, 3, 2, 1, 1), so for 11 counters guess the 4, 3, 2, 1 stairs plus one extra: $5, 3, 2, 1, 4, 4, 2, 1, 4, 3, 3, 1, 4, 3, 2, 2, 4, 3, 2, 1, 1$; running $5, 3, 2, 1 \rightarrow 4, 4, 2, 1 \rightarrow 4, 3, 3, 1 \rightarrow 4, 3, 2, 2 \rightarrow 4, 3, 2, 1, 1 \rightarrow 5, 3, 2, 1$ confirms it. (3) 15 counters from one pile: $15 \rightarrow 14, 1 \rightarrow 13, 2 \rightarrow 12, 2, 1 \rightarrow 11, 3, 1 \rightarrow 10, 3, 2 \rightarrow 9, 3, 2, 1 \rightarrow 8, 4, 2, 1 \rightarrow 7, 4, 3, 1 \rightarrow 6, 4, 3, 2 \rightarrow 5, 4, 3, 2, 1$, 10 moves.

Grades 4–5 table (three children, one adult)

This table runs the same experiments and then argues. Let the children run games in parallel and pool results; insist that rows are written biggest first with commas, and that “the loop” is named by listing its arrangements. The run tables have two or more spare rows, so the number of rows never gives away the number of moves.

Order and timing (40 minutes): Problems 1–2 (6 min), 3 (6; the list of 11 and why it is complete), 4 (5), 5–6 (6), 7 (4), 8 (4; the first argument), 9 (3), 10 (1), 11 (2), 12 (3; the second argument). Problems 13–17, on the last three pages, are for a table that is ahead, in any order the children like; 14 and 15 are the best of them, and 13 can be shared out.

Answers, grades 4–5

- **Problem 1:** $2, 2, 1, 1 \rightarrow 4, 1, 1 \rightarrow 3, 3 \rightarrow 2, 2, 2 \rightarrow 3, 1, 1, 1 \rightarrow 4, 2 \rightarrow 3, 2, 1 \rightarrow 3, 2, 1$; stopped after 6 moves.
- **Problem 2:** $3, 3 \rightarrow 2, 2, 2 \rightarrow 3, 1, 1, 1 \rightarrow 4, 2 \rightarrow 3, 2, 1$ (4 moves); $6 \rightarrow 5, 1 \rightarrow 4, 2 \rightarrow 3, 2, 1$ (3); $1, 1, 1, 1, 1, 1 \rightarrow 6 \rightarrow \dots$ (4). All stop at $3, 2, 1$.
- **Problem 3:** the 11 arrangements, biggest first pile first, with the moves until the game stops in brackets: 6 (3), $5, 1$ (2), $4, 2$ (1), $4, 1, 1$ (5), $3, 3$ (4), $3, 2, 1$ (0), $3, 1, 1, 1$ (2), $2, 2, 2$ (3), $2, 2, 1, 1$ (6), $2, 1, 1, 1, 1$ (3), $1, 1, 1, 1, 1, 1$ (4). Why the list is complete: sort by the biggest pile. Biggest pile 6: one way. Biggest 5: the other 1 counter is a pile of 1, one way. Biggest 4: the other 2 counters are 2 or 1, 1, two ways. Biggest 3: the other 3 counters are $3, 2, 1$ or $1, 1, 1$, three ways. Biggest 2: the other 4 counters in piles of at most 2 are $2, 2, 2, 1, 1$ or $1, 1, 1, 1, 1$, three ways. Biggest 1: one way. $1 + 1 + 2 + 3 + 3 + 1 = 11$. Yes, every start ends at $3, 2, 1$, and 6 moves is the most it can take. Say this explicitly: because the list is complete, it is a proof, not a collection of examples.
- **Problem 4:** $4 \rightarrow 3, 1 \rightarrow 2, 2 \rightarrow 2, 1, 1 \rightarrow 3, 1$ (row 4 repeats row 1: goes round on three arrangements). $5 \rightarrow 4, 1 \rightarrow 3, 2 \rightarrow 2, 2, 1 \rightarrow 3, 1, 1 \rightarrow 3, 2$ (row 5 repeats row 2: goes round on three). The seven arrangements of 5: $5; 4, 1; 3, 2; 3, 1, 1; 2, 2, 1; 2, 1, 1, 1; 1, 1, 1, 1, 1$. The argument: the rows Start, 1, 2, $\dots$, 7 are eight arrangements, but only seven exist, so two rows are equal, and the later of the two is a repeat, at move 7 at the latest. Then push: once a row repeats an earlier row, everything after it repeats too, because each row is determined by the row above it. That is why every game, with any number of counters, either stops or goes round.
- **Problem 5:** $8 \rightarrow 7, 1 \rightarrow 6, 2 \rightarrow 5, 2, 1 \rightarrow 4, 3, 1 \rightarrow 3, 3, 2 \rightarrow 3, 2, 2, 1 \rightarrow 4, 2, 1, 1 \rightarrow 4, 3, 1$. Row 8 repeats row 4: it goes round, and the loop is $4, 3, 1, 3, 3, 2, 3, 2, 2, 1, 4, 2, 1, 1$: four arrangements.
- **Problem 6:** $2, 2, 2, 2 \rightarrow 4, 1, 1, 1, 1 \rightarrow 5, 3 \rightarrow 4, 2, 2 \rightarrow 3, 3, 1, 1 \rightarrow 4, 2, 2$: a loop of two, $4, 2, 2$ and $3, 3, 1, 1$. $5, 1, 1, 1 \rightarrow 4, 4 \rightarrow 3, 3, 2 \rightarrow 3, 2, 2, 1 \rightarrow 4, 2, 1, 1 \rightarrow 4, 3, 1 \rightarrow 3, 3, 2$: the same loop of four as the game from 8. So the games from 8 and from $5, 1, 1, 1$ share a loop; the game from $2, 2, 2, 2$ does not.
- **Problem 7:** $10 \rightarrow 9, 1 \rightarrow 8, 2 \rightarrow 7, 2, 1 \rightarrow 6, 3, 1 \rightarrow 5, 3, 2 \rightarrow 4, 3, 2, 1$ (stopped after 6 moves); $5, 5 \rightarrow 4, 4, 2 \rightarrow 3, 3, 3, 1 \rightarrow 4, 2, 2, 2 \rightarrow 4, 3, 1, 1, 1 \rightarrow 5, 3, 2 \rightarrow 4, 3, 2, 1$ (6); $2, 2, 2, 2, 2 \rightarrow 5, 1, 1, 1, 1, 1 \rightarrow 6, 4 \rightarrow 5, 3, 2 \rightarrow 4, 3, 2, 1$ (4). All three stop at the stairs $4, 3, 2, 1$.
- **Problem 8:** $4, 3, 2, 1$ does not change, and it is the only arrangement of 10 counters that does not. There is no such arrangement of 8 counters. The argument, in steps the children can follow: (a) if the move changes nothing, the new pile must be one of the piles you see afterwards, and it is the biggest one, because every other pile just lost a counter and was at most the old biggest. So the biggest pile

- **Problem 9:** 1, 1, 1 $\rightarrow$ 3 $\rightarrow$ 2, 1 $\rightarrow$ 2, 1 (stopped after 2). 7 $\rightarrow$ 6, 1 $\rightarrow$ 5, 2 $\rightarrow$ 4, 2, 1 $\rightarrow$ 3, 3, 1 $\rightarrow$ 3, 2, 2 $\rightarrow$ 3, 2, 1, 1 $\rightarrow$ 4, 2, 1 (row 7 repeats row 3; loop of four). 5, 4 $\rightarrow$ 4, 3, 2 $\rightarrow$ 3, 3, 2, 1 $\rightarrow$ 4, 2, 2, 1 $\rightarrow$ 4, 3, 1, 1 $\rightarrow$ 4, 3, 2 (row 5 repeats row 1; loop of four: 4, 3, 2, 3, 3, 2, 1, 4, 2, 2, 1, 4, 3, 1, 1). Table: 3 stops; 4 goes round, one loop of 3; 5 goes round, one loop of 3; 6 stops; 7 goes round, a loop of 4; 8 goes round, loops of 4 and of 2; 9 goes round, a loop of 4; 10 stops. For 7 and 9 the children have seen one loop each; in fact it is the only one (all 15 arrangements of 7 and all 30 of 9 reach it; tables at the end). The stopping numbers are 3, 6, 10.
- **Problem 10:** the next stairs, 5, 4, 3, 2, 1, has 15 counters; then 21 and 28, with stairs 6, 5, 4, 3, 2, 1 and 7, 6, 5, 4, 3, 2, 1. If a child wants to test 15 at once: 3, 3, 3, 3, 3 $\rightarrow$ 5, 2, 2, 2, 2, 2 $\rightarrow$ 6, 4, 1, 1, 1, 1, 1 $\rightarrow$ 7, 5, 3 $\rightarrow$ 6, 4, 3, 2 $\rightarrow$ 5, 4, 3, 2, 1, stopped after 5 moves (every child has 21 counters).
- **Problem 11** (undoing a move). The method: the new pile after a move equals the number of piles before it; so choose which pile is “new”, give every other pile one counter back, and add piles of 1 until the number of piles equals the new pile. Into 3, 2, 1: from 4, 2 (new pile 2: the other piles 3, 1 came from 4, 2, and 2 piles is right) and from 3, 2, 1 itself (new pile 3: others 2, 1 came from 3, 2, plus one pile of 1). Into 5, 3, 2: from 6, 4 (new pile 2), from 6, 3, 1 (new pile 3: others 5, 2 came from 6, 3, plus one pile of 1), from 4, 3, 1, 1, 1 (new pile 5: others 3, 2 came from 4, 3, plus three piles of 1). Into 2, 2, 2, 2: nothing, and the reason is the point: the new pile would have to be a 2, so there were 2 piles before the move, but three other piles remain after it, and a move cannot create piles except the new one. Impossible.
- **Problem 12:** 4, 4 is reached from 5, 1, 1, 1; 3, 3, 1, 1 from 4, 2, 2; 5, 1, 1, 1 from 2, 2, 2, 1, 1; 4, 2, 1, 1 from 3, 2, 2, 1. Unreachable: 2, 2, 2, 2, 2, 2, 2, 1, 1, 3, 2, 1, 1, 1, 3, 1, 1, 1, 1, 1. What they have in common: the biggest pile is smaller than the number of piles minus one. The reason: after a move, the new pile is as big as the number of piles before the move, and that number is at least the number of piles after the move minus one (all the piles you see afterwards, except the new one, were there before). So after any move, the biggest pile is at least the number of piles minus one. Test: 6, 1, 1 has biggest pile 6 and 3 piles, so it can be reached (from 2, 2, 1, 1, 1, 1, new pile 6); 2, 2, 2, 1, 1, 1, 1 has biggest pile 2 and 7 piles, so it cannot. Checked against the list of 22 in Problem 13, the rule marks exactly seven as unreachable: 3, 2, 1, 1, 1, 3, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 1, 1, 2, 2, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 (the starred rows of the 8-counter table at the end). Arrangements that cannot be reached are called Garden of Eden states.
- **Problem 13:** S (the short loop): 6, 1, 1, 5, 3, 4, 2, 2, 4, 1, 1, 1, 1, 3, 3, 1, 1, 2, 2, 2, 2, 2, 2, 1, 1, 1, 1, seven arrangements. L (the long loop): the other fifteen. Each run is in the 8-counter table at the end.
- **Problem 14:** the slowest starts with 10 counters take 12 moves: 3, 3, 2, 1, 1, 3, 2, 2, 2, 1, 3, 2, 2, 1, 1, 1. For example 3, 3, 2, 1, 1 $\rightarrow$ 5, 2, 2, 1 $\rightarrow$ 4, 4, 1, 1 $\rightarrow$ 4, 3, 3 $\rightarrow$ 3, 3, 2, 2 $\rightarrow$ 4, 2, 2, 1, 1 $\rightarrow$ 5, 3, 1, 1 $\rightarrow$ 4, 4, 2 $\rightarrow$ 3, 3, 3, 1 $\rightarrow$ 4, 2, 2, 2 $\rightarrow$ 4, 3, 1, 1, 1 $\rightarrow$ 5, 3, 2 $\rightarrow$ 4, 3, 2, 1. Eleven moves: 6, 2, 1, 1, 5, 2, 2, 1, 5, 2, 1, 1, 1. The number of moves for all 42 arrangements is in the table at the end. A theorem (Igusa 1985, Etienne 1991) says that with $1 + 2 + \dots + k$ counters no start needs more than $k^2 - k$ moves: 12 for 10 counters, 20 for 15, 30 for 21.

- **Problem 15:** the five arrangements are 5, 3, 2, 1, 4, 4, 2, 1, 4, 3, 3, 1, 4, 3, 2, 2, 4, 3, 2, 1, 1, and 5, 3, 2, 1 $\rightarrow$ 4, 4, 2, 1 $\rightarrow$ 4, 3, 3, 1 $\rightarrow$ 4, 3, 2, 2 $\rightarrow$ 4, 3, 2, 1, 1 $\rightarrow$ 5, 3, 2, 1 stays inside the list: the extra counter walks down the staircase one step per move and then reappears on top. For 8 counters, the six arrangements with one extra counter on each of two different steps of 3, 2, 1 are 4, 3, 1, 4, 2, 2, 4, 2, 1, 1, 3, 3, 2, 3, 3, 1, 1, 3, 2, 2, 1, which are exactly the four arrangements of the long loop together with the two of the short loop. In the long loop the two extras sit on neighboring steps and walk down together; in the short loop they sit two steps apart.
- **Problem 16:** the ten arrangements are 5, 4, 2, 1, 5, 3, 3, 1, 5, 3, 2, 2, 5, 3, 2, 1, 1, 4, 4, 3, 1, 4, 4, 2, 2, 4, 4, 2, 1, 1, 4, 3, 3, 2, 4, 3, 3, 1, 1, 4, 3, 2, 2, 1. Two loops of five: 5, 4, 2, 1 $\rightarrow$ 4, 4, 3, 1 $\rightarrow$ 4, 3, 3, 2 $\rightarrow$ 4, 3, 2, 2, 1 $\rightarrow$ 5, 3, 2, 1, 1 $\rightarrow$ 5, 4, 2, 1 (extras on neighboring steps) and 5, 3, 3, 1 $\rightarrow$ 4, 4, 2, 2 $\rightarrow$ 4, 3, 3, 1, 1 $\rightarrow$ 5, 3, 2, 2 $\rightarrow$ 4, 4, 2, 1, 1 $\rightarrow$ 5, 3, 3, 1 (extras two steps apart). Not the same loop. Starts outside the list that end in the loop with 5, 3, 3, 1: 6, 4, 2 (one move), 7, 5 (two), 3, 3, 3, 3 (four), 2, 2, 2, 2, 2, 2 (four), 4, 4, 4 (five: 4, 4, 4 $\rightarrow$ 3, 3, 3, 3 $\rightarrow$ 4, 2, 2, 2, 2 $\rightarrow$ 5, 3, 1, 1, 1, 1 $\rightarrow$ 6, 4, 2 $\rightarrow$ 5, 3, 3, 1). Of the 77 arrangements of 12 counters, 45 end in the loop with 5, 4, 2, 1 and 32 in the loop with 5, 3, 3, 1.
- **Problem 17:** from 5, 5, 5 it is 17 moves: 5, 5, 5 $\rightarrow$ 4, 4, 4, 3 $\rightarrow$ 4, 3, 3, 3, 2 $\rightarrow$ 5, 3, 2, 2, 2, 1 $\rightarrow$ 6, 4, 2, 1, 1, 1 $\rightarrow$ 6, 5, 3, 1 $\rightarrow$ 5, 4, 4, 2 $\rightarrow$ 4, 4, 3, 3, 1 $\rightarrow$ 5, 3, 3, 2, 2 $\rightarrow$ 5, 4, 2, 2, 1, 1 $\rightarrow$ 6, 4, 3, 1, 1 $\rightarrow$ 5, 5, 3, 2 $\rightarrow$ 4, 4, 4, 2, 1 $\rightarrow$ 5, 3, 3, 3, 1 $\rightarrow$ 5, 4, 2, 2, 2 $\rightarrow$ 5, 4, 3, 1, 1, 1 $\rightarrow$ 6, 4, 3, 2 $\rightarrow$ 5, 4, 3, 2, 1. From 4, 4, 3, 2, 1, 1 it is 20 moves, the most possible with 15 counters: 4, 4, 3, 2, 1, 1 $\rightarrow$ 6, 3, 3, 2, 1 $\rightarrow$ 5, 5, 2, 2, 1 $\rightarrow$ 5, 4, 4, 1, 1 $\rightarrow$ 5, 4, 3, 3 $\rightarrow$ 4, 4, 3, 2, 2 $\rightarrow$ 5, 3, 3, 2, 1, 1 $\rightarrow$ 6, 4, 2, 2, 1 $\rightarrow$ 5, 5, 3, 1, 1 $\rightarrow$ 5, 4, 4, 2 $\rightarrow$ 4, 4, 3, 3, 1 $\rightarrow$ 5, 3, 3, 2, 2 $\rightarrow$ 5, 4, 2, 2, 1, 1 $\rightarrow$ 6, 4, 3, 1, 1 $\rightarrow$ 5, 5, 3, 2 $\rightarrow$ 4, 4, 4, 2, 1 $\rightarrow$ 5, 3, 3, 3, 1 $\rightarrow$ 5, 4, 2, 2, 2 $\rightarrow$ 5, 4, 3, 1, 1, 1 $\rightarrow$ 6, 4, 3, 2 $\rightarrow$ 5, 4, 3, 2, 1. The two runs merge at 5, 4, 4, 2.

Hints, grades 4–5, in the order to give them

1. Wrong rows (usually a forgotten pile): “The new pile equals the number of piles before the move. Check that first.”
2. Problem 3, finding the list: “What is the biggest pile you can have? With a biggest pile of 4, what can the rest be?” Then “with a biggest pile of 3, the rest is 3 counters in piles of at most 3; how many ways?” Then “is anything missing between the ones you have?” For “how do you know?”: “Say the rule you used to go from one line to the next.”
3. Problem 4: “How many rows have you written by move 7? How many different arrangements exist?”
4. Problem 8: “After the move, which pile is the new one? Could it be smaller than some other pile?” then “Cover the biggest pile with your hand. What is left, before and after?”
5. Problem 11: “The new pile tells you how many piles there were. Choose which pile is new.”
6. Problem 12: “Compare the biggest pile with the number of piles in each unreachable one.”
7. Problem 14: “The slowest start for 6 counters was 2, 2, 1, 1: small piles, but not all ones. Try something like that with 10.”
8. Problem 15: “Write the loop for 7 counters as stairs 3, 2, 1, 0 with one extra counter each, and watch where the extra counter goes at each move.”

If the grades 4–5 table stalls or finishes early

Stalls: Problem 3 can be cut to making the list and checking it is complete (read them the number of moves for the ones they have not run, from the 6-counter table at the end). Problem 9 can be cut to the table only, with you supplying the runs for 7 and 9. If the arguments in Problems 8 and 12 do not come, do them as a conversation with counters on the table: make a move, cover the new pile, ask what is left.

Finishes early: Problems 13–17 (14 and 15 are the best; 13 can be shared out; 17 is long). After that: (1) 21 counters in one pile takes 15 moves ($21 \rightarrow 20, 1 \rightarrow 19, 2 \rightarrow 18, 2, 1 \rightarrow 17, 3, 1 \rightarrow 16, 3, 2 \rightarrow 15, 3, 2, 1 \rightarrow 14, 4, 2, 1 \rightarrow 13, 4, 3, 1 \rightarrow 12, 4, 3, 2 \rightarrow 11, 4, 3, 2, 1 \rightarrow 10, 5, 3, 2, 1 \rightarrow 9, 5, 4, 2, 1 \rightarrow 8, 5, 4, 3, 1 \rightarrow 7, 5, 4, 3, 2 \rightarrow 6, 5, 4, 3, 2, 1$); from 5, 5, 4, 3, 2, 1, 1 it takes 30, the most possible. (2) Predict the loops for 9 counters (the 3, 2, 1 stairs with extras on three of the four steps: 4, 3, 2, 4, 3, 1, 1, 4, 2, 2, 1, 3, 3, 2, 1, one loop of four) and for 13 (stairs 4, 3, 2, 1 with extras on three of five steps: ten arrangements, two loops of five). (3) Count the Garden of Eden arrangements for 6, 7, 8 counters using the rule from Problem 12 (3, 5, 7) and check against the lists.

Share (5 minutes)

Each table shows one thing, with counters, in this order. K–1: the three stairs they built (3, 6, 10), one child holding each or one child with all three, and the claim “these numbers stop; 4 goes round and round” (and 5, 7, 8 if they played them). Grades 2–3: the map of 6 counters, held up, with the sentence “every arrow chain ends at the stairs,” and the 8-counter surprise (two different loops). Grades 4–5: why no arrangement of 8 counters can stand still, in the three steps of Problem 8, and the complete list of 11 arrangements of 6 from Problem 3, with the sentence “we did not check some starts, we checked all of them.” Close with the question nobody has answered yet: “we checked every arrangement of 6 counters by hand, and a computer checked all 42 arrangements of 10. How would you know it for 15, 21, 28, without checking?” That is the theorem.

The mathematics behind the session (for the two mathematicians)

Setting

A state is a partition of n ; write it as towers of dots, columns of heights $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$ standing on a floor. The move B removes the bottom row (one dot from each of the m columns) and stands it up as a new column of height m ; then the columns are re-sorted by height. The state space is finite and B is a function, so every orbit is eventually periodic: this is the pigeonhole argument of grades 4–5 Problem 4, and the functional graph (the “map” of grades 2–3 Problem 9) consists of cycles with trees hanging off them.

Fixed points. If $B(\lambda) = \lambda$ then the new column, of height m , must be the tallest column afterwards (every other column shrank), so $\lambda_1 = m$. Removing that column from both sides, the multiset $\{\lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_m - 1\}$ with zeros dropped must equal $\{\lambda_2, \dots, \lambda_m\}$; comparing largest elements repeatedly gives $\lambda_2 = \lambda_1 - 1$, $\lambda_3 = \lambda_2 - 1$, and so on down to $\lambda_m = 1$. So the fixed points are exactly the staircases $(k, k - 1, \dots, 1)$, and only the triangular numbers $T_k = k(k + 1)/2$ admit one (grades 4–5 Problem 8).

Brandt’s theorem and the structure of the cycles

Theorem (Brandt 1982). If $n = T_k$, every state reaches the staircase. More precisely, for every n with $T_k \leq n < T_{k+1}$, write $n = T_k + r$ with $0 \leq r \leq k$; the states that lie on cycles are exactly the staircase

$(k, \dots, 1)$ with one extra dot added to each of r distinct “steps” (a step being one of the k columns or a new column of height 1), and B moves the set of occupied steps cyclically: if the extras occupy steps $S \subseteq \{0, 1, \dots, k\}$ (step j meaning column of height j gets one more; step 0 meaning a new column of height 1), then after a move they occupy $S - 1$ modulo $k + 1$. Hence the cycle lengths divide $k + 1$, the number of cycle states is $\binom{k+1}{r}$, and the number of cycles is the number of binary necklaces of length $k + 1$ with r ones. For $n = T_k$ ($r = 0$) the only cycle is the fixed point.

Checks against the session: $n = 8 = T_3 + 2$, necklaces of length 4 with two ones: 1100 (period 4) and 1010 (period 2), the loops L and S; $n = 12 = T_4 + 2$, necklaces of length 5 with two ones: two of them, both of period 5; $n = 7, 9, 11, 14$: one cycle each; $n = 17 = T_5 + 2$: three cycles, of lengths 6, 6 and 3.

Proof sketch (a potential function plus a rotation argument). Number the cells of a state by row r (from the floor, $r = 1$ at the bottom) and column c (from the left). Perform the move in two steps. Step 1: remove the bottom row and put it as a new column at the far left, shifting the old columns one place right; a surviving cell goes from (r, c) to $(r - 1, c + 1)$ and a bottom cell from $(1, c)$ to $(c, 1)$. Every cell keeps its diagonal number $d = r + c$. Step 2: if the new column list $(m, \lambda_1 - 1, \dots, \lambda_m - 1)$ is not in descending order, sort it. Let $W(\lambda) = \sum_{\text{cells}} (r + c)$. Step 1 keeps W ; step 2 permutes column heights, and $\sum_c c \cdot h_c$ is minimized, strictly, by sorting the heights in descending order (rearrangement inequality), so W strictly decreases whenever sorting is needed. Since W is a positive integer, sorting happens only finitely often along any orbit; in particular, on a cycle, no move ever needs sorting, and every cell simply travels along its diagonal: on diagonal d the $d - 1$ positions $(1, d - 1), (2, d - 2), \dots, (d - 1, 1)$ are visited cyclically, with period $d - 1$.

Now suppose a cycle state has a hole (an empty position) on diagonal d while some cell sits on diagonal $d + 1$. The hole rotates with period $d - 1$ and the cell with period d ; these are coprime, so at some time in the cycle the hole and the cell are in the same row, the cell at $(r, d + 1 - r)$ immediately to the right of the empty $(r, d - r)$. That contradicts descending column heights. Hence on a cycle the diagonals 2, 3, $\dots$ are full, full, $\dots$, full, then possibly one partially filled diagonal, then empty. The full diagonals form a staircase T_k , the partial diagonal $k + 2$ carries the r extras on distinct steps, and the rotation is the cyclic shift described above. If $n = T_k$ the partial diagonal is empty and the cycle is the fixed point. (Brandt’s original proof is different; this route is close to Akin and Davis, 1985.)

How many moves

Igusa (1985) and Etienne (1991) proved that for $n = T_k$ every start reaches the staircase within $k^2 - k$ moves. The bound is attained by $(k - 1, k - 1, k - 2, \dots, 2, 1, 1)$ for $k = 3, 4, 5, 6$: 6 moves from 2, 2, 1, 1, 12 from 3, 3, 2, 1, 1, 20 from 4, 4, 3, 2, 1, 1, 30 from 5, 5, 4, 3, 2, 1, 1 (every fact in this paragraph was checked by exhaustive computation while preparing this guide). For 10 counters the distribution of the number of moves over the 42 arrangements is: 0 moves, 1 arrangement; 1, 1; 2, 3; 3, 5; 4, 5; 5, 3; 6, 4; 7, 4; 8, 4; 9, 3; 10, 3; 11, 3; 12, 3.

Garden of Eden states

A state with m piles has a predecessor if and only if its largest pile is at least $m - 1$: choosing a pile of size $p \geq m - 1$ as the new pile, the other $m - 1$ piles came from piles one larger, and $p - (m - 1) \geq 0$ piles of size 1 complete the predecessor. So the Garden of Eden states are the partitions with largest part at most (number of parts) $- 2$; by conjugation these are counted by partitions of n with largest part minus number of parts at least 2, that is, Dyson rank at least 2, which is the route by which Hopkins and Sellers (2007) enumerated them. Counts for $n = 3, \dots, 15$: 1, 1, 2, 3, 5, 7, 10, 14, 20, 27, 37, 49, 66. Undoing a move (grades 4–5 Problem 11) is the same computation read backwards; a state can have several predecessors (3, 2, 1 has two, 5, 3, 2 has three).

Where it leads

Undergraduate: partitions and Young diagrams; functional graphs and eventually periodic dynamics on finite sets; the rearrangement inequality as a potential-function tool; necklaces and Burnside counting for the number of cycles; Dyson's rank. Research: the transient length and the shape of the trees (Griggs and Ho 1998 study the analogous shift on compositions and other "shift" dynamics); generalizations such as Austrian, Carolina and Montreal solitaire and σ -shifts on partitions (surveyed by Hopkins 2012); random Bulgarian solitaire, where each pile contributes a counter with probability p , studied as a Markov chain; the exact enumeration of Garden of Eden partitions and the distribution of the number of moves.

References. J. Brandt, Cycles of partitions, Proc. Amer. Math. Soc. 85 (1982). M. Gardner, Mathematical Games, Scientific American, August 1983 (the column that made the game known). E. Akin and M. Davis, Bulgarian solitaire, Amer. Math. Monthly 92 (1985). K. Igusa, Solution of the Bulgarian solitaire conjecture, Math. Magazine 58 (1985). G. Etienne, Tableaux de Young et solitaire bulgare, J. Combin. Theory Ser. A 58 (1991). J. R. Griggs and C.-C. Ho, The cycling of partitions and compositions under repeated shifts, Adv. in Appl. Math. 21 (1998). B. Hopkins and J. A. Sellers, Exact enumeration of Garden of Eden partitions, Integers 7(2) (2007). B. Hopkins, 30 years of Bulgarian solitaire, College Math. J. 43 (2012).

Tables: every arrangement, its successor, and its distance

Each table lists every arrangement (Start), the arrangement after one move (Next), the number of moves until the game stops or first enters its loop (Moves), and, for 8 counters, which loop it enters (L: the loop of four, S: the loop of two). A star in the GoE column marks a Garden of Eden arrangement, one that no move produces.

6 counters, 11 arrangements

Start	Next	Moves	GoE
6	5, 1	3	
5, 1	4, 2	2	
4, 2	3, 2, 1	1	
4, 1, 1	3, 3	5	
3, 3	2, 2, 2	4	
3, 2, 1	3, 2, 1	0	
3, 1, 1, 1	4, 2	2	
2, 2, 2	3, 1, 1, 1	3	
2, 2, 1, 1	4, 1, 1	6	*
2, 1, 1, 1, 1	5, 1	3	*
1, 1, 1, 1, 1, 1	6	4	*

7 counters, 15 arrangements

Start	Next	Moves	GoE
7	6, 1	3	
6, 1	5, 2	2	
5, 2	4, 2, 1	1	
5, 1, 1	4, 3	2	
4, 3	3, 2, 2	1	
4, 2, 1	3, 3, 1	0	
4, 1, 1, 1	4, 3	2	
3, 3, 1	3, 2, 2	0	
3, 2, 2	3, 2, 1, 1	0	
3, 2, 1, 1	4, 2, 1	0	
3, 1, 1, 1, 1	5, 2	2	*
2, 2, 2, 1	4, 1, 1, 1	3	*
2, 2, 1, 1, 1	5, 1, 1	3	*
2, 1, 1, 1, 1, 1	6, 1	3	*
1, 1, 1, 1, 1, 1, 1	7	4	*

8 counters, 22 arrangements

Start	Next	Moves	Loop	GoE
8	7, 1	4	L	
7, 1	6, 2	3	L	
6, 2	5, 2, 1	2	L	
6, 1, 1	5, 3	2	S	
5, 3	4, 2, 2	1	S	
5, 2, 1	4, 3, 1	1	L	
5, 1, 1, 1	4, 4	2	L	
4, 4	3, 3, 2	1	L	
4, 3, 1	3, 3, 2	0	L	
4, 2, 2	3, 3, 1, 1	0	S	
4, 2, 1, 1	4, 3, 1	0	L	
4, 1, 1, 1, 1	5, 3	2	S	
3, 3, 2	3, 2, 2, 1	0	L	
3, 3, 1, 1	4, 2, 2	0	S	
3, 2, 2, 1	4, 2, 1, 1	0	L	
3, 2, 1, 1, 1	5, 2, 1	2	L	*
3, 1, 1, 1, 1, 1	6, 2	3	L	*
2, 2, 2, 2	4, 1, 1, 1, 1	3	S	*
2, 2, 2, 1, 1	5, 1, 1, 1	3	L	*
2, 2, 1, 1, 1, 1	6, 1, 1	3	S	*
2, 1, 1, 1, 1, 1, 1	7, 1	4	L	*
1, 1, 1, 1, 1, 1, 1, 1	8	5	L	*

9 counters, 30 arrangements (1 of 2)

Start	Next	Moves	GoE
9	8, 1	5	
8, 1	7, 2	4	
7, 2	6, 2, 1	3	
7, 1, 1	6, 3	3	
6, 3	5, 2, 2	2	
6, 2, 1	5, 3, 1	2	
6, 1, 1, 1	5, 4	2	
5, 4	4, 3, 2	1	
5, 3, 1	4, 3, 2	1	
5, 2, 2	4, 3, 1, 1	1	
5, 2, 1, 1	4, 4, 1	6	
5, 1, 1, 1, 1	5, 4	2	
4, 4, 1	3, 3, 3	5	
4, 3, 2	3, 3, 2, 1	0	
4, 3, 1, 1	4, 3, 2	0	

9 counters, 30 arrangements (2 of 2)

Start	Next	Moves	GoE
4, 2, 2, 1	4, 3, 1, 1	0	
4, 2, 1, 1, 1	5, 3, 1	2	
4, 1, 1, 1, 1, 1	6, 3	3	*
3, 3, 3	3, 2, 2, 2	4	
3, 3, 2, 1	4, 2, 2, 1	0	
3, 3, 1, 1, 1	5, 2, 2	2	*
3, 2, 2, 2	4, 2, 1, 1, 1	3	
3, 2, 2, 1, 1	5, 2, 1, 1	7	*
3, 2, 1, 1, 1, 1	6, 2, 1	3	*
3, 1, 1, 1, 1, 1, 1	7, 2	4	*
2, 2, 2, 2, 1	5, 1, 1, 1, 1	3	*
2, 2, 2, 1, 1, 1	6, 1, 1, 1	3	*
2, 2, 1, 1, 1, 1, 1	7, 1, 1	4	*
2, 1, 1, 1, 1, 1, 1, 1	8, 1	5	*
1, 1, 1, 1, 1, 1, 1, 1, 1	9	6	*

10 counters, 42 arrangements (1 of 2)

Start	Next	Moves	GoE
10	9, 1	6	
9, 1	8, 2	5	
8, 2	7, 2, 1	4	
8, 1, 1	7, 3	9	
7, 3	6, 2, 2	8	
7, 2, 1	6, 3, 1	3	
7, 1, 1, 1	6, 4	3	
6, 4	5, 3, 2	2	
6, 3, 1	5, 3, 2	2	
6, 2, 2	5, 3, 1, 1	7	
6, 2, 1, 1	5, 4, 1	11	
6, 1, 1, 1, 1	5, 5	7	
5, 5	4, 4, 2	6	
5, 4, 1	4, 3, 3	10	
5, 3, 2	4, 3, 2, 1	1	
5, 3, 1, 1	4, 4, 2	6	
5, 2, 2, 1	4, 4, 1, 1	11	
5, 2, 1, 1, 1	5, 4, 1	11	
5, 1, 1, 1, 1, 1	6, 4	3	
4, 4, 2	3, 3, 3, 1	5	
4, 4, 1, 1	4, 3, 3	10	

10 counters, 42 arrangements (2 of 2)

Start	Next	Moves	GoE
4, 3, 3	3, 3, 2, 2	9	
4, 3, 2, 1	4, 3, 2, 1	0	
4, 3, 1, 1, 1	5, 3, 2	2	
4, 2, 2, 2	4, 3, 1, 1, 1	3	
4, 2, 2, 1, 1	5, 3, 1, 1	7	
4, 2, 1, 1, 1, 1	6, 3, 1	3	*
4, 1, 1, 1, 1, 1, 1	7, 3	9	*
3, 3, 3, 1	4, 2, 2, 2	4	
3, 3, 2, 2	4, 2, 2, 1, 1	8	
3, 3, 2, 1, 1	5, 2, 2, 1	12	*
3, 3, 1, 1, 1, 1	6, 2, 2	8	*
3, 2, 2, 2, 1	5, 2, 1, 1, 1	12	*
3, 2, 2, 1, 1, 1	6, 2, 1, 1	12	*
3, 2, 1, 1, 1, 1, 1	7, 2, 1	4	*
3, 1, 1, 1, 1, 1, 1, 1	8, 2	5	*
2, 2, 2, 2, 2	5, 1, 1, 1, 1, 1	4	*
2, 2, 2, 2, 1, 1	6, 1, 1, 1, 1	8	*
2, 2, 2, 1, 1, 1, 1	7, 1, 1, 1	4	*
2, 2, 1, 1, 1, 1, 1, 1	8, 1, 1	10	*
2, 1, 1, 1, 1, 1, 1, 1, 1	9, 1	6	*
1, 1, 1, 1, 1, 1, 1, 1, 1, 1	10	7	*